1. The data set used in Exercise 5.3 is used again. This time it is used to estimate how the proportion of the household budget spent on transportation WTRANS depends on the log of total expenditure ln(TOTEXP), AGE, and number of children NK. The output is reported in Table 5.7.

(a) Write out the estimated equation in the standard reporting format with standard errors below the coefficient estimates.

(b) Interpret the estimates b_2, b_3, and b_4. Do you think the results make sense from an economic or logical point of view?

(c) Are there any variables that you might exclude from the equation? Why?

(d) What proportion of variation in the budget proportion allocated to transport is explained by this equation?

(e) Predict the proportion of a budget that will be spent on transportation, for both one- and two-children households, when total expenditure and age are set at their sample means, which are 98.7 and 36, respectively.

Table 5.7 **Output for Exercise 5.4**

Dependent Variable: *WTRANS*
Included observations: 1519

Variable	Coefficient	Std. Error	*t*-Statistic	Prob.
C	−0.0315	0.0322	−0.9776	0.3284
ln(*TOTEXP*)	0.0414	0.0071	5.8561	0.0000
AGE	−0.0001	0.0004	−0.1650	0.8690
NK	−0.0130	0.0055	−2.3542	0.0187
R-squared	0.0247		Mean dependent var	0.1323
			S.D. dependent var	0.1053

2. This question is concerned with the value of houses in towns surrounding Boston. It uses the data of Harrison, D., and D. L. Rubinfeld (1978), ‘‘Hedonic Prices and the Demand for Clean Air,’’ Journal of Environmental Economics and Management, 5, 81–102. The output appears in Table 5.8. The variables are defined as follows:

VALUE = median value of owner-occupied homes in thousands of dollars

CRIME = per capita crime rate

NITOX = nitric oxide concentration (parts per million)

ROOMS = average number of rooms per dwelling

AGE = proportion of owner-occupied units built prior to 1940

DIST = weighted distances to five Boston employment centers

ACCESS = index of accessibility to radial highways

TAX = full-value property-tax rate per $10,000

PTRATIO = pupil–teacher ratio by town

(a) Report briefly on how each of the variables influences the value of a home.

(b) Find 95% interval estimates for the coefficients of CRIME and ACCESS.

(c) Test the hypothesis that increasing the number of rooms by one increases the value of a house by $7,000.

(d) Test as an alternative hypothesis H_1 that reducing the pupil–teacher ratio by 10 will increase the value of a house by more than $10,000.

Table 5.8 **Output for Exercise 5.5**

Dependent Variable: *VALUE*
Included observations: 506

Variable	Coefficient	Std. Error	*t*-Statistic	Prob.
C	28.4067	5.3659	5.2939	0.0000
CRIME	−0.1834	0.0365	−5.0275	0.0000
NITOX	−22.8109	4.1607	−5.4824	0.0000
ROOMS	6.3715	0.3924	16.2378	0.0000
AGE	−0.0478	0.0141	−3.3861	0.0008
DIST	−1.3353	0.2001	−6.6714	0.0000
ACCESS	0.2723	0.0723	3.7673	0.0002
TAX	−0.0126	0.0038	−3.3399	0.0009
PTRATIO	−1.1768	0.1394	−8.4409	0.0000

3. Suppose that from a sample of 63 observations, the least squares estimates and the corresponding estimated covariance matrix are given by

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}, \quad \widehat{\operatorname{cov}(b)} = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Test each of the following hypotheses and state the conclusion:

(a) $\beta_2 = 0$

(b) $\beta_1 + 2\beta_2 = 5$

(c) $\beta_1 - \beta_2 + \beta_3 = 4$

Homework

4. The file ***cocaine.csv*** contains 56 observations on variables related to sales of cocaine powder in northeastern California over the period 1984–1991. The data are a subset of those used in the study Caulkins, J. P. and R. Padman (1993), ‘‘Quantity Discounts and Quality Premia for Illicit Drugs,’’ Journal of the American Statistical Association, 88, 748–757. The variables are

PRICE = price per gram in dollars for a cocaine sale

QUANT = number of grams of cocaine in a given sale

QUAL = quality of the cocaine expressed as percentage purity

TREND = a time variable with 1984 = 1 up to 1991 = 8

Consider the regression model

$$PRICE = \beta_1 + \beta_2 QUANT + \beta_3 QUAL + \beta_4 TREND + e$$

(a) What signs would you expect on the coefficients β_2, β_3, and β_4?

(b) Use your computer software to estimate the equation. Report the results and interpret the coefficient estimates. Have the signs turned out as you expected?

(c) What proportion of variation in cocaine price is explained jointly by variation in quantity, quality, and time?

(d) It is claimed that the greater the number of sales, the higher the risk of getting caught. Thus, sellers are willing to accept a lower price if they can make sales in larger quantities. Set up H_0 and H_1 that would be appropriate to test this hypothesis. Carry out the hypothesis test.

(e) Test the hypothesis that the quality of cocaine has no influence on price against the alternative that a premium is paid for better-quality cocaine.

(f) What is the average annual change in the cocaine price? Can you suggest why price might be changing in this direction?

5. The file ***br2.csv*** contains data on 1,080 houses sold in Baton Rouge, Louisiana, duringmid-2005.We will be concerned with the selling price (PRICE), the size of the house in square feet (SQFT), and the age of the house in years (AGE).

(a) Use all observations to estimate the following regression model and report the results

$$PRICE = \beta_1 + \beta_2 SQFT + \beta_3 AGE + e$$

(i) Interpret the coefficient estimates.

(ii) Find a 95% interval estimate for the price increase for an extra square foot of living space—that is, $\partial PRICE / \partial SQFT$.

(iii) Test the hypothesis that having a house a year older decreases price by 1000 or less $\left(H_0 : \beta_3 \geq -1000\right)$ against the alternative that it decreases price by more than 1000 $\left(H_1 : \beta_3 < -1000\right)$.

(b) Add the variables $SQFT^2$ and AGE^2 to the model in part (a) and re-estimate the equation. Report the results.

(i) Find estimates of the marginal effect $\partial PRICE / \partial SQFT$ for the smallest house in the sample, the largest house in the sample, and a house with 2300 SQFT. Comment on these values. Are they realistic?

(ii) Find estimates of the marginal effect $\partial \text{PRICE} / \partial \text{AGE}$ for the oldest house in the sample, the newest house in the sample, and a house that is 20 years old. Comment on these values. Are they realistic?

(iii) Find a 95% interval estimate for the marginal effect $\partial \text{PRICE} / \partial \text{SQFT}$ for a house with 2300 square feet.

(iv) For a house that is 20 years old, test the hypothesis

$$\text{H}_0 : \frac{\partial \text{PRICE}}{\partial \text{AGE}} \geq -1000 \text{ against } \text{H}_1 : \frac{\partial \text{PRICE}}{\partial \text{AGE}} < -1000$$

(c) Add the interaction variable SQFT$\times$ AGE to the model in part (b) and reestimate the equation. Report the results. Repeat parts (i), (ii), (iii), and (iv) from part (b) for this new model. Use SQFT = 2300 and AGE = 20.

(d) From your answers to parts (a), (b), and (c), comment on the sensitivity of the results to the model specification.